A Dialogue on Artificial Intelligence

Leonardo Pisano (Fibonacci)

with

Fibonacci: Salutations. I am Leonardo of Pisa, whom some call Fibonacci. I once brought the Hindu-Arabic numerals to the merchants of Italy, for numbers are tools that change what minds can do. You speak of “artificial intelligence”; pray, tell me, what craft is this?

You: It is a family of methods that let machines learn patterns from vast examples. Today’s most capable systems are neural networks, especially a design called transformers, which generate text, code, and images.

Fibonacci: Ah—machines that learn by many examples as an apprentice watching masters. In my day, I learned methods from the Arabs and applied them to the accounts of merchants. Your machines learn from ledgers many orders thicker than any library I knew. What is this “transformer”?

You: It attends to all parts of a sentence at once, weighing context to predict the next word. By repeating this, it composes paragraphs, answers questions, and even writes proofs at a sketch level.

Fibonacci: Weighing every word against every other—like a careful merchant who, before concluding a price, reviews all entries in the ledger. In Liber Abaci

I taught practical reckoning so that trade might flow. Your device seems a universal reckoner for language. Does it reason as we do?

You: Partly. It excels at patterns, analogies, and composing coherent text. But it can be unreliable, mixing truth with invention unless constrained or checked by formal tools.

Fibonacci: Then it is as a very swift apprentice—astonishingly productive yet needing a master's eye. You speak of formal tools; do you mean the Euclidean style of certainty?

You: Yes. We have proof assistants that mechanize logic, so a computer can check each step. Some are building bridges between generative AI and formal proof, so the machine drafts and the assistant verifies.

Fibonacci: This pleases me. Commerce prized speed, but geometry prized certainty. If your art joins swift draft with strict verification, you unite the virtues of the market and the academy. Tell me, how do you instruct such machines? With rules, or with numbers?

You: With numbers. Every word, image, or sound becomes a vector of numbers. The model adjusts millions or billions of parameters by gradient descent so predictions improve over time.

Fibonacci: A most numerical rhetoric! When I replaced the counting board with written numerals, calculation became supple; new methods bloomed. Representation governs possibility. You have found a representation of meaning in

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number, and now calculation touches language itself.

You: Exactly. Your famous rabbits gave us a simple recurrence—the sequence now bearing your name. Modern models also depend on recurrences, attention patterns, and scaling laws linking data, parameters, and performance.

Fibonacci: Then a lesson persists: from a clear rule and abundant input, growth follows. Yet, growth without measure courts folly. Who ensures these engines do no harm?

You: We develop safeguards: alignment, audits, provenance tracking, and laws for data and deployment. There is debate about open versus closed models, privacy, and the future of work.

Fibonacci: In my travels I saw that methods spread through trust—of merchants, teachers, rulers. So too with your engines: reputation, standard, and governance must walk together. What of mathematics itselfâ€”do these minds discover, or only rearrange?

You: They rediscover known results and sketch new connections, but true novelty still often requires human insight. However, they accelerate search and suggest promising paths, which we can then formalize.

Fibonacci: A compass, not a captain—yet a fine compass multiplies voyages. You, being a mathematician, speak of foundations and structures beyond fields, even “hyperfields.” Would such formal novelties suit these engines?

You: Yes. We can encode definitions, examples, and theorems for machines to explore. They can propose conjectures in algebraic and logical settings, and we guide them toward proofs within assistant systems.

Fibonacci: Then the scholar's role is elevated, not erased. As I taught algorismus to merchants so they might reckon with ease, you may teach these engines the language of proof so that scholars reason farther. Still, remember the merchant's caution: verify the coin, not only the count.

You: Wise counsel. If you were here today, what would you build?

Fibonacci: I would compile a new Liber—of data and demonstrations. First, a **primer of representations:** how words, images, and numbers are encoded. Next, **a book of trustworthy practices:** draft by the machine, certify by proof. Lastly, **problems of the commonwealth:** medicine, trade, and learning, each set with examples the machine may study and the human may judge. Thus the art would be popular and prudent.

You: And what would you ask of us?

Fibonacci: Three things. Teach the young to speak with machines as they do with numbers. Wed invention to verification, so speed does not outrun truth. And keep the channels of learning open, that this new arithmetic of thought may enrich all cities, not a few.

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